# Shadowplay

A lamp on your desk, a story on your wall. Your hands cast the characters.

Entertainment track | New Experience | Target: Best New Entertainment Experience

## AT A GLANCE

| | |
|---|---|
| **Track / division** | Entertainment (immersive storytelling) / New Experience |
| **Primary prize** | Best New Entertainment Experience ($100,000; runner-up $50,000) |
| **Fallback special award** | Best First Five Minutes ($25,000) |
| **Session length** | One chapter is 5 to 8 minutes; the full story is about 35 minutes over 5 chapters |
| **Room requirement** | A wall in front of the seated player, or a floating screen fallback |
| **Core tech** | WebXR hand tracking, plane detection (wall), real-time shadow mapping, silhouette matching |
| **Biggest risk** | Recognising hand shapes reliably, especially when both hands overlap |
| **Why it wins attention** | Everyone already knows how to make a shadow rabbit: zero learning curve, instant delight |
| **Availability assumed** | Full time: roughly 300 to 400 hours through 17 Nov |

### The one-line pitch

An interactive shadow-puppet story performed on your real wall. A virtual lamp on your desk throws the shadows of your real hands across the room; form the right shape and your shadow comes alive as a character, then carries the story forward.

SECTION 01

# The idea

Shadowplay is a short interactive story told entirely in shadows on the wall of your own room. You sit at your desk. A small warm lamp appears on the desk in front of you and the room dims slightly, like a theatre before a show. When you raise your hands between the lamp and the wall, their shadows appear on your real wall, stretched and soft, exactly as they would with a real lamp.

The story asks you for characters. A narrator says the field is too wide for a moth to cross alone, and a faint outline of a rabbit appears on the wall. You make the rabbit shape with your hand. When your shadow matches, it comes to life: the ears twitch, it blinks, and it hops across the wall when you move your hand. From then on you are both the audience and the puppeteer.

### Why it fits the Entertainment track

The track asks for lean-back media and immersive storytelling with a hands-first implementation. Shadowplay is lean-back in pace (narrated, gentle, no fail states) but the story literally cannot move without the player's hands. It is also one of the oldest forms of storytelling there is, reinvented with technology that can see your fingers.

SECTION 02

# The core insight

Shadow puppetry is a hand language almost everyone already speaks. Most people learned a rabbit, a dog or a bird as children. That solves the hardest problem in controller-free design, which is teaching gestures: the player arrives already knowing the vocabulary.

The second insight is that light obeys physics people understand. Move your hand toward the lamp and the shadow grows; move it toward the wall and it shrinks and sharpens. Shadowplay turns that into a puzzle mechanic, so the player's real-world intuition drives the story.

| Physical rule | How it becomes gameplay |
|---|---|
| **Closer to the lamp means a bigger shadow** | Grow your wolf big enough to scare off the storm, or shrink a mouse small enough to slip through a keyhole. |
| **Shadows overlap and merge** | Two hands together form one larger creature, such as a bird with two wings. |
| **Moving the lamp moves every shadow** | Pinch and drag the lamp to sweep a searchlight across the wall and reveal hidden things. |
| **Soft versus sharp edges** | A sharp, close shadow is 'solid' and can push objects; a soft, far one is a ghost that can pass through them. |

SECTION 03

# The story: The Moth Who Borrowed Light

An original story written for this project (placeholder title). A small moth has lost its glow to the Long Night and must cross the world on your wall to borrow light from the lighthouse at the edge of the sea. Each chapter introduces a new shape and a new use of the light.

| Chapter | Setting and beat | Shapes and mechanics introduced | Length |
|---|---|---|---|
| **Prologue** | The lamp flickers on; the moth lands on its shade and tells its problem. | Open hand (wave hello); the moth follows your fingertip. | 3 min |
| **1. The Field** | Tall grass the moth cannot cross. | Rabbit: hop across; move your hand to steer. | 6 min |
| **2. The River** | A dark river under the moon. | Fish or boat; distance to the lamp controls size, so the boat must be big enough to carry the moth. | 7 min |
| **3. The Wolf** | A shadow wolf blocks the path. | Dog or wolf; choose: grow huge and scare it, or become a small bird and fly past. A branching choice with two endings for this chapter. | 7 min |
| **4. The Storm** | Wind and rain on the wall. | Drag the lamp to find shelter; two-hand bird to fly. | 6 min |
| **5. The Lighthouse** | The keeper is asleep; the light is out. | Cupped hands hold the light; open them to release it across the whole wall. Finale. | 6 min |

Each chapter ends on a complete moment, so the one bus stop test holds: a player can do one chapter in a short break and feel finished.

SECTION 04

# The player experience

## The first five minutes

| Time | What happens |
|---|---|
| **0:00** | Passthrough fades in. A small lamp appears on the desk; the room dims gently except for a pool of warm light on the wall ahead. |
| **0:15** | The player's hands naturally enter the light. Their shadows appear on the wall. Most people start making shapes immediately, unprompted. |
| **0:40** | A moth flutters onto the lampshade. A narrator (voice plus English captions) says hello. The moth flies to the player's shadow fingertip and follows it. |
| **1:30** | The first request: a faint rabbit outline appears on the wall. The player makes the rabbit shape; the outline fills in and the shadow comes alive. |
| **2:30** | The rabbit hops wherever the hand moves. The player carries the moth across the field. |

| Time | What happens |
|---|---|
| **4:30** | Chapter 1 ends. The rabbit is added to the Shadow Journal, a small book on the desk that collects every creature discovered. |

### Interaction vocabulary

| Interaction | How the player does it | Effect |
|---|---|---|
| **Cast a shape** | Form a hand shape in the lamp light and hold it for half a second. | Matching shape comes alive as a character. |
| **Puppeteer** | Move, tilt or open and close the hand. | The character moves; opening and closing fingers moves its mouth or wings. |
| **Resize** | Move the hand toward or away from the lamp. | The character grows or shrinks. |
| **Move the lamp** | Pinch the lamp and drag it along the desk. | Every shadow and the light pool shift across the wall. |
| **Release** | Open the hand fully and pull it out of the light. | The character bows and fades; the hand is free again. |
| **Pause** | Turn the lamp off with a pinch on its switch, or take the headset off. | The story pauses; captions show where you left off. |

SECTION 05

## Why passthrough and the real room matter

- **Your wall is the stage.** The story is projected onto the real wall in front of you, at the real size of that wall, lit by a lamp on your real desk. With passthrough off there is no wall and no theatre.
- **Your real hands cast the shadows.** The characters are made from your own hands, which you can see in passthrough at the same moment as their shadows.
- **The room dims like a theatre.** A WebXR app cannot dim passthrough directly, so the app draws a large translucent dark shell around the player with a soft-edged opening where the stage is. Passthrough shows through it darker everywhere except the light pool: a moment of ceremony that turns an ordinary room into a stage.
- **The stage adapts to the wall.** The light pool and the story's scene fit the detected wall size and distance. A close wall gives an intimate show; a far one gives a big-screen show.

### Holding up in rooms the developer never tested

The app looks for the wall plane most directly in front of the seated player, between about 0.6 and 4 metres away. **The most common case is a desk pushed against a wall**, so the wall is often only 0.6 to 1 metre away and partly blocked by a monitor. For close walls, the stage sits higher on the wall, above the monitor, and is scaled down; the lamp moves closer to the player so shadows stay large enough. If no suitable wall is found (the player faces an open room or a window), it places a softly glowing paper screen floating 2 metres ahead. The story plays identically in all three cases, so no room blocks the experience.

SECTION 06

## Seated, FoV-aware design

- The stage is about 1.2 m wide at a distance of about 2 m, roughly 33 degrees across, which fits inside the Quest 3S field of view without head turning. On a close wall behind a monitor (about 0.8 m), it shrinks to about 0.5 m wide to keep the same angle.
- Hands work in a comfortable zone just above the desk, between the lamp and the wall. Elbows can rest on the desk.
- Captions sit at the bottom of the stage, never at the edges of the view.
- No standing or walking. The whole show happens from a chair.

SECTION 07

## Shape recognition: the hard part

Shadowplay uses two complementary methods to decide whether the player's hand matches a target shape.

### Method 1: joint-angle templates

WebXR hand tracking reports 25 joints per hand. For each frame the app computes a compact feature vector: how curled each finger is, how far apart the fingers are spread, where the thumb tip is relative to the other fingers, and which way the palm faces relative to the lamp. It compares this vector with stored templates for each shape and accepts a match within a tolerance. This is cheap enough to run every frame.

### Method 2: silhouette matching

Because what matters is the shadow, not the exact hand pose, the app can also compare shadows directly. A few times per second it renders the hands' shadow into a tiny 64 x 64 black-and-white image and measures how much it overlaps the target silhouette. A match is accepted when the overlap is above a threshold (starting around 55%) for half a second. This is forgiving in exactly the right way: any hand pose that looks like a rabbit counts as a rabbit, even if the fingers are arranged differently.

> **Design rule: choose shapes that hand tracking can see**
>
> Hand tracking degrades when fingers are hidden behind other fingers or when two hands overlap or interlock. The story is therefore built mainly on **one-hand shapes** (rabbit, dog, duck, snail, deer, fish) and on two-hand shapes where the hands sit **side by side** rather than interlocked (the bird uses two hands next to each other as wings). Every two-hand shape also has a one-hand alternative for accessibility.

SECTION 08

## Progression and reasons to come back

| System | Description |
|---|---|
| **Chapters** | Five chapters, each a complete moment. Players return to see what happens next. |
| **The Shadow Journal** | A small book on the desk that collects every creature discovered, with a page showing its shape. Hidden creatures (swan, snail, crocodile, deer) can be found by experimenting. |
| **Shape of the Day** | A daily riddle ('I carry my house on my back') with a new shape to discover. Solving it adds a page to the journal. |
| **Free Stage** | After the story, an open stage mode where any discovered creature can be performed freely with music, for play or to show friends. |
| **Encore endings** | Chapter 3's branching choice gives a reason to replay. |

SECTION 09

## Accessibility

- Adjustable recognition tolerance, from strict to very forgiving.
- A one-hand alternative for every two-hand shape.
- Full English captions for all narration, placed on the stage.
- No time pressure and no fail states; the story waits for the player.
- A 'show me' option that briefly displays a ghost hand demonstrating the shape, so no one gets stuck.
- Optional voice alternative: if speech recognition is available in Quest Browser, saying the creature's name also summons it (needs verification on device).

SECTION 10

## Art direction and sound

### Look

Warm lamplight on a real wall, deep soft shadows, and characters drawn as elegant silhouettes with small animated details (a blinking eye, fluttering ears). The palette is amber, ink black and moonlight blue. Silhouette art is forgiving and quick to make, which suits a solo schedule: every character is a 2D shape with a few animated parts.

### Sound

- A music-box and soft strings score, **commissioned or properly licensed**. Self-composed music is only competitive if you already produce music.
- Narration by a **commissioned voice actor**, credited in the submission. Text-to-speech only with a licence that allows commercial, competition and promotional use.
- **Licensing for a web app is stricter.** The audio files are publicly downloadable from the hosted site, so licences must allow distributing them, not just playing them. Use work-for-hire agreements for commissions and keep every licence and release in the repo.
- Send the voice and music briefs in week 1: commissioned work takes weeks to turn around.
- Spatial audio: sounds come from the wall where the character is, and the lamp hums faintly from the desk.
- Each creature has a small signature sound when it comes alive.

SECTION 11

## Technical architecture (IWSDK)

| Feature | Used for |
|---|---|
| **Hand tracking** | Joint poses for shape recognition; a simple hand mesh built from joints casts the shadows. |
| **Plane detection** | Finding the wall in front of the player (semantic label 'wall'). |
| **Shadow mapping** | One spot light (the lamp) with a 1024 x 1024 shadow map; the wall is a transparent surface that only shows shadows. |
| **Render-to-texture** | The 64 x 64 silhouette buffer for matching, updated a few times per second rather than every frame. |
| **Anchors** | Keeping the lamp and stage fixed in the room. |
| **Web Audio / spatial audio** | Narration, score and positioned sound effects. |

| System | Responsibility |
|---|---|
| **StageSetupSystem** | Chooses the wall or the floating screen fallback; places the lamp on the desk. |
| **HandCasterSystem** | Updates the hand shadow meshes from joint poses every frame. |
| **ShapeMatchSystem** | Runs joint-angle matching each frame and silhouette matching at about 5 Hz; emits 'shape matched' events. |
| **PuppetSystem** | Once matched, maps hand motion to character motion and animation. |
| **StorySystem** | A state machine driven by a JSON script of beats, requests and branches. |
| **NarrationSystem** | Plays narration and shows captions in sync. |
| **JournalSystem** | Records discovered creatures in IndexedDB. |

### Performance notes

- One shadow-casting light only; keep the shadow map at 1024 pixels.
- Reading pixels back from the GPU is slow and can stall a frame, so the silhouette check runs on a tiny buffer at a low rate using asynchronous readback (a WebGL2 pixel buffer plus a fence) where available, and joint-angle matching carries most of the work.
- WebXR reports hand joints but no hand mesh. The shadow caster is either simple capsules along the joints or a skinned hand model bundled with the app (never loaded from an external CDN at runtime). The week-1 spike decides which looks convincing.
- Characters are flat textured quads with a few bones, so drawing cost is tiny.

SECTION 12

## Risks and mitigations

| Risk | Severity | Mitigation |
|---|---|---|
| **Shape recognition is unreliable** | High | Two methods combined; generous tolerances; shapes chosen for trackability; 'show me' helper. Tested in week 1. |
| **Two-hand shapes break tracking** | High | Avoid interlocking shapes; side-by-side only; one-hand alternatives. |
| **The story is not compelling** | Medium | Write the full script in week 1; test it as a read-aloud with friends before building. |
| **Narration and music quality** | Medium | Commission a voice actor and composer with distribution rights; keep the score simple. |
| **No wall in front of the player** | Low | Floating paper screen fallback. |
| **Shadow rendering cost** | Low | Single light; small shadow map; tested early on device. |
| **The wall is right behind the monitor** | High | The most common desk setup. Stage placed above the monitor and scaled; lamp moved closer; tested in the playtest rooms. |
| **Joint-capsule shadows look like sausages, not hands** | Medium | Week-1 spike compares capsules with a bundled skinned hand model; pick whichever reads as a real hand shadow. |
| **Silhouette readback stalls frames** | Medium | Tiny buffer, low rate, asynchronous readback; joint templates as the primary method. |
| **Passthrough cannot be captured for the video** | High | Verify on day 1 that Quest recording or casting captures passthrough with the WebXR layer; if not, plan an alternative before week 4. |
| **Tracking quality varies with lighting and skin tone** | Medium | Playtest across varied lighting, hand sizes and skin tones; tune recognition on the whole set, not the developer's hands. |
| **Commissioned audio arrives late** | Medium | Brief the voice actor and composer in week 1; build with placeholder audio; final audio due by week 5. |

## Week 1: spikes

More hours reduce scope and content risk, but not platform risk: hand-tracking accuracy and how convincing a rendered shadow looks do not improve with effort. Week 1 measures those before the story is built.

| Day | Spike | What to measure |
|---|---|---|
| **1** | Passthrough capture check (first hour) | Record a short clip of a WebXR scene over passthrough using Quest recording or casting. If it fails, the video plan changes. |
| **1 - 2** | Shadow render | Cast hand shadows from a lamp onto the detected wall, comparing joint capsules with a bundled skinned hand model. Judge on device whether they read as real hand shadows. Test a close wall behind a monitor and a far wall. |
| **2 - 4** | Shape recognition | Record templates for 8 one-hand shapes and 2 side-by-side two-hand shapes. Measure recognition accuracy across at least 5 people with different hand sizes and skin tones, for joint templates alone and with silhouette matching added. |
| **5** | Story and audio | Full script table read with two or three listeners; send the voice-actor and composer briefs. |

### Go criteria for Shadowplay

Shadows read as real hand shadows on device, on both a close and a far wall. At least 90% recognition for the one-hand shapes across all testers, with no false matches while hands are simply moving through the light. Listeners want to know what happens after the prologue.

## Decision order

1. **All pass:** build Shadowplay at full scope.
2. **Recognition below target for some shapes:** drop or replace those shapes and loosen tolerances (about 1 day). The story is written around shapes, so swapping one is cheap in week 1.
3. **Two-hand shapes unreliable:** make every two-hand moment optional, with its one-hand alternative as the default.
4. **Shadows do not look convincing with either method:** this is the concept's core, so reconsider the idea choice before building further.

SECTION 13

# Seven-week build plan

| Week | Dates | Goal | Exit criterion |
|---|---|---|---|
| **Week 1** | 29 Sep - 4 Oct | Spikes: capture, shadow render, recognition across 5+ people; script table read; commission voice and music | Go criteria met; capture method confirmed; briefs sent |
| **Week 2** | 5 - 11 Oct | Stage setup (close wall, far wall, floating fallback); room dimming; lamp grab; recognition system; puppeteering | A matched rabbit hops around the wall under hand control in all three stage cases |
| **Week 3** | 12 - 18 Oct | Story engine (JSON beats); Prologue and Chapter 1 with placeholder audio; first five minutes | First 10 minutes playable end to end |
| **Week 4** | 19 - 25 Oct | Chapters 2 and 3 including the branch; Shadow Journal; **start capturing video footage** | Half the story complete; first usable footage |
| **Week 5** | 26 Oct - 1 Nov | Chapters 4 and 5; final narration and score integrated; Shape of the Day; Free Stage | Content-complete for the must-ship list |
| **Week 6** | 2 - 8 Nov | Playtests in 10+ rooms (lighting, skin tones, hand sizes, desk-against-wall, no wall); accessibility options; performance pass | 72 fps held; no tester gets stuck; every room gets a stage |
| **Week 7** | 9 - 15 Nov | First-five-minutes iteration until every cold tester succeeds; animation and sound polish; final video edit; submission text | Video cut from weeks of footage; build frozen candidate |
| **Ship** | 16 - 18 Nov | Final checks on a clean device; submit on 17 Nov | Submitted a day early; hosted link verified |

### Where the extra full-time hours go

The content load (script, voice direction, five chapters) now fits the schedule. Beyond that, spend hours where judges score: **playtests** in 10 or more real rooms, **iterating the first five minutes** until every cold tester makes their first shape unprompted, and **video footage** captured from week 4 onward.

SECTION 14

## Scope

| Must ship | Nice to have | Cut |
|---|---|---|
| **Wall detection with close-wall handling and floating fallback; room dimming; lamp; 8 to 10 shapes; all five chapters including the branching choice; commissioned narration and score; captions; Shadow Journal; Shape of the Day; Free Stage; tolerance, one-hand and 'show me' options** | Voice alternative for summoning creatures (if speech recognition works in Quest Browser); more hidden creatures | Multiplayer shadow theatre, user-authored stories and accounts: a backend adds failure modes during the judging window and adds nothing to the score. In-app video recording (the Quest's own capture covers it). |

SECTION 15

## Demo video plan (under 3 minutes)

| Time | Beat | Shot |
|---|---|---|
| **0:00 - 0:12** | Hook | A dim room, a lamp clicks on, a hand rises and a shadow rabbit on the real wall blinks and hops away. |
| **0:12 - 0:40** | Setup | Wide shot showing the real wall, desk and lamp; the room dimming like a theatre. |
| **0:40 - 1:30** | Play | Real-time Chapter 1: the moth, the rabbit request, the shape match, puppeteering. |
| **1:30 - 2:10** | Physics as puzzle | Moving the hand to the lamp to grow the wolf; dragging the lamp to reveal a hidden path. |
| **2:10 - 2:40** | Discovery | Flipping through the Shadow Journal; a hidden snail found by experimenting. |
| **2:40 - 2:55** | Close | Finale light spreading across the wall; title. |

SECTION 16

## Submission draft

| | |
|---|---|
| **Name** | Shadowplay |
| **Tagline (under 140 characters)** | An interactive shadow-puppet story on your real wall. Your hands cast the characters, and the story waits for them. |
| **Track / division** | Entertainment / New Experience |
| **Target launch** | Web release Q1 2027 with additional chapters; Horizon Store PWA after that |

### Description (first draft)

Almost everyone learned to make a shadow rabbit as a child. Shadowplay turns that shared memory into an interactive story told on the wall of your own room.

A small lamp appears on your real desk and the room dims like a theatre. Your hands cast soft shadows on your real wall. When the story needs a character, you make its shape, and your shadow comes alive: a rabbit that hops where your hand goes, a wolf that grows as you move closer to the lamp, a bird made of two hands. You help a small moth cross the world to borrow light from a distant lighthouse.

We built Shadowplay with the Immersive Web SDK. Plane detection finds the wall in front of you, a real-time shadow map casts your tracked hands onto it, and two recognition methods, joint-angle templates and silhouette matching, decide when your shadow matches a character. There are no controllers, no menus to learn and no text walls: the first shape you make is the tutorial.

SECTION 17

## How it maps to the judging criteria

| Criterion (25% each) | Evidence in Shadowplay |
|---|---|
| **Innovation and Creativity** | An ancient art form reinvented with hand tracking; light physics as a puzzle mechanic. |
| **Experience Design** | Pre-learned gestures, no text tutorials, a room that becomes a theatre, chapters sized for short sessions. |
| **Technical Implementation** | Wall detection with fallback, real-time shadows from tracked hands, dual recognition methods. |
| **Polish and Presentation** | Silhouette art is striking and achievable solo; a hook that reads instantly on video. |

### Hard requirement and design guidelines

| Test | What the rules ask | How this idea passes |
|---|---|---|
| **Hands-first (hard requirement)** | Complete the entire experience without ever pairing a controller. | Every interaction is a hand shape, a hand movement or a pinch on the lamp. |
| **Airplane seat test** | Every interaction works within a two-foot radius, seated. | Hands work in the space just above the desk; the stage is directly ahead. |
| **One bus stop test** | Fast cold start, clean pause/resume, a satisfying moment in 10 minutes or less. | Each chapter is 5 to 8 minutes and complete; pausing is a lamp switch or removing the headset. |
| **Take it away test** | Remove any third-party service and a project still stands. | No third-party services; all content is original and runs locally. |

### Competition rules that shape this plan

- **One prize per entry.** Rule 7 limits each entry to a single prize, so this brief aims at one track prize and names one special award as a fallback rather than trying to stack awards.
- **Submission:** a public HTTPS link (GitHub Pages, Vercel or similar) that stays live until the winner announcement (about 11 Dec 2026), a public YouTube/Vimeo video under 3 minutes, and the Devpost form (name, 140-character tagline, track, division, description of about 500 words, hand-interaction notes, target launch date).
- **No changes after the deadline.** Judging uses the build as it stood at 18 Nov 2026, 12:00 PT. Plan to submit on 17 Nov and freeze the hosted build.
- **Content rules:** English (or English subtitles), no identifiable real people other than entrants, no brand names or logos, all music, art and fonts properly licensed or original.
- **Video rules:** real footage from a Quest or the XR simulator; do not lean on AI-generated video to carry the pitch.
- **Performance floor:** judges expect at least 60 fps on Quest hardware. Target 72 fps.
- **Voice and music:** keep licences or signed releases for narration and score.

SECTION 18

## Honest weaknesses

- Recognition quality decides everything. If shapes are frustrating to trigger, the magic disappears.
- Storytelling quality is harder to guarantee than mechanics; a weak script makes it feel like a tech demo.
- Replay value after the story ends is limited; the journal, daily shape and Free Stage help, but it is a story first.
- Narration and music take real money and lead time; commission them in week 1.
- The desk-against-wall case makes the stage smaller and closer than the ideal; it must still feel like a theatre.

SECTION 19

## Scorecard and verdict

| Dimension | Score / 10 | Comment |
|---|---|---|
| **Originality** | 9 | Distinctive and instantly understood. |
| **Only possible in MR** | 8 | Needs the real wall and real hands; the fallback screen weakens it slightly. |
| **Works in untested rooms** | 8 | One wall, or the fallback. |
| **Solo feasibility in 7 weeks** | 8 | Full-time hours cover the story content; recognition is the remaining hard part. |
| **Prize ceiling** | $100k | Top Entertainment prize; First Five Minutes as a fallback. |

### Verdict

The best backup to Tabletop Beat, and a strong first choice on its own if you prefer storytelling to games. With full-time hours its content load (script, voice, five chapters) fits the schedule, so the full story can ship. It has the most emotional pull of the six ideas and the strongest first five minutes.

SECTION 20

# Build status (as of 2 Oct 2026)

Shadowplay is built on IWSDK (WebXR, Three.js) in the **4KInc/shadowplay** repository. All work is on the branch **feat/shadowplay-chapters-1-3**, open for review as **pull request #1** into main. The whole story, from the Prologue to the lighthouse finale, plus Free Stage, the Shadow Journal, Shape of the Day and the accessibility options, is playable end to end in the IWER emulator (Meta Quest 3 profile, living-room scene) with emulated hand input. On-device testing on a real Quest has not started yet.

## Must-ship list (Section 14) against what is built

| Must-ship item | Status | What exists |
|---|---|---|
| **Wall detection, close-wall handling, floating fallback** | **Built** | Picks the wall straight ahead (0.6 to 4 m) whose detected outline contains the point ahead; close walls lift and shrink the stage above the monitor; narrow walls get a smaller stage; paper-screen fallback; re-places on headset recenter or from Settings. |
| **Room dimming** | **Built** | Translucent shell with a soft opening onto the stage, fades in like a theatre; opens up in the finale. |
| **Lamp** | **Built** | Procedural desk lamp, one spot light with a 1024 px soft shadow map. Pinch-drag along the desk sweeps the light pool across the wall; pinch the switch to pause. |
| **8 to 10 shapes** | **Built** | Eleven: rabbit (plus an easier "ears" variant), fish, dog, bird (two hands), cupped hands, and the hidden snail, swan, deer (two hands) and crocodile (two hands); open hand for continue and release. |
| **All five chapters incl. the branching choice** | **Built** | Prologue and Chapters 1 to 5 as a 102-beat JSON script: moth follows the fingertip, rabbit carries her, fish must grow big enough, wolf choice with two endings, storm search with the lamp, held light and finale. |
| **Commissioned narration and score** | **Pipeline ready** | NarrationSystem plays per-beat clips from a manifest and the story waits for each line. No recordings yet: placeholder synthesised sound effects and an optional on-device test voice. |
| **Captions** | **Built** | Every line captioned at the bottom of the stage, cross-faded, readable on any wall. |
| **Shadow Journal** | **Built** | Desk book, pinch to open; eight creature pages; endings count; saved in IndexedDB. |
| **Shape of the Day** | **Built** | Date-seeded riddle, solve remembered per day, adds the creature to the journal. |
| **Free Stage** | **Built** | After the story: any known creature comes alive, with a placeholder music-box loop. |
| **Tolerance, one-hand and show-me options** | **Built** | Hand-ray Settings panel beside the journal: tolerance, hint timing, one-hand mode, placeholder voice, chapter select, re-place stage, replay story. |

## Technical architecture (Section 11) against what is built

| Brief | Status | Implementation |
|---|---|---|
| **Hand tracking** | **Built** | Joint poses each frame; capsule shadow casters built from the 25 joints of each hand. |
| **Plane detection** | **Built** | Vertical planes labelled wall; outline containment and fitting as above. |
| **Shadow mapping** | **Built** | One spot light at the bulb, 1024 px PCF soft shadows; wall surface shows only shadows. |
| **Render-to-texture silhouettes** | **Built** | Each hand rendered from the lamp into 64 x 64 at 5 Hz, read back asynchronously (WebGL2 pixel buffer and fence), normalised to 32 x 32 and compared by overlap. |
| **Anchors** | **Partly** | Not using XRAnchor yet; instead the stage re-places itself when the reference space resets. |
| **Web Audio / spatial audio** | **Built** | HRTF-panned placeholder sounds: chimes from the wall, clicks and a lamp hum from the desk. |
| **Systems** | **Built** | StageSetup, HandCaster, ShapeMatch, Puppet, Story, Narration, Journal as planned, plus Silhouette, LampControl, Moth, StageActor (wolf), Scenery, HiddenProp, Caption, FreeStage, Settings and Audio. |

SECTION 21

# Where we are against the seven-week plan

| Week | Planned goal (Section 13) | Status |
|---|---|---|
| **Week 1**<br>**29 Sep - 4 Oct** | Spikes: capture, shadow render, recognition across 5+ people, script table read, commission voice and music | **Partly** Shadow rendering and both recognition methods are built and work in the emulator. The device spikes (passthrough capture, real-hand accuracy across testers), the table read and the commissions have not happened. |
| **Week 2**<br>**5 - 11 Oct** | Stage setup in all three cases, room dimming, lamp grab, recognition, puppeteering | **Built** In the emulator; the close-wall case is implemented but not yet seen on a real desk. |
| **Week 3**<br>**12 - 18 Oct** | Story engine, Prologue and Chapter 1, first five minutes | **Built** |
| **Week 4**<br>**19 - 25 Oct** | Chapters 2 and 3 with the branch, Shadow Journal, start capturing footage | **Built** Footage capture not started. |
| **Week 5**<br>**26 Oct - 1 Nov** | Chapters 4 and 5, final narration and score, Shape of the Day, Free Stage | **Built** Except the real narration and score, which need the commissioned recordings. |
| **Week 6**<br>**2 - 8 Nov** | Playtests in 10+ rooms, accessibility options, performance pass | **Partly** Accessibility options done; desktop performance pass done; playtests and on-device performance not started. |
| **Week 7**<br>**9 - 15 Nov** | First-five-minutes iteration, polish, final video, submission text | **Partly** Desktop title card and hands-first guard done; iteration with cold testers and the video need a headset. |

### Ahead on content, behind on device validation

Content planned for weeks 2 to 6 was built during week 1. But the risks week 1 was meant to measure are still open: whether the shadows read as real hands on a real wall, whether real hands reach 90% recognition across testers, and whether passthrough can be captured for the video. None of these can be settled in an emulator. The decision order in Section 12 still applies, and the next hours should go on a Quest rather than on more features.

SECTION 22

# Verification evidence and known gaps

Checked in the emulator with emulated hand input (typecheck and production build pass after every slice):

- Stage: wall chosen by outline containment (a wall section 1.2 m to the side is correctly rejected); fallback screen when facing open space; Re-place from Settings.
- Shadows cast from tracked hands; every chapter path played, including both Chapter Three endings, the storm search by sliding the lamp and the finale.
- Lamp pinch-drag and switch pause; journal pinch; settings, chapter select and replay pressed with hand rays; journal, settings, endings and progress survive a reload and a dev-server restart.
- Silhouette matching: same pose 100% overlap, same pose moved closer to the lamp 89%, hand rolled 90 degrees 43% (rejected); a pose the joint method misreads still summons the rabbit from its shadow alone.
- Narration: a test clip held the story until it ended; placeholder voice speaks every line; desktop title card; controllers-only sessions pause until hands return.
- Desktop frame timing p50 8.3 ms, p95 10.2 ms, no dropped frames; 50 draw calls and about 41k triangles in normal play.

## Known gaps

| Gap | Why it is open | How it closes |
|---|---|---|
| **Real-hand recognition of rabbit, fish, dog, cup, snail, swan, crocodile** | The emulator can only make an open hand and a pinch; those requests were tested through the accessibility alternative | Week-1 recognition spike on a Quest with 5+ people of different hand sizes and skin tones |
| **No silhouette templates shipped** | Templates must be recorded from real hands | Record with the built-in dev recorder, export to public/silhouettes/templates.json |
| **72 fps on Quest 3 / 3S unmeasured** | Only desktop profiling so far | On-device performance trace, then fixes |
| **Shadow look on a real wall** | Capsule casters judged only in the emulator | Compare with a skinned hand model on close and far walls (Week-1 spike) |
| **Spatial audio, narration, recenter re-placement** | The emulator plays no audio and cannot trigger a recenter | Headset check |

| Gap | Why it is open | How it closes |
|---|---|---|
| **Passthrough capture for the video** | Not tested | Day-one capture spike |
| **Public HTTPS link** | Hosting not set up (needs approval to publish) | Deploy (e.g. Vercel) and keep live until the winner announcement |
| **Commissioned voice and score** | Not commissioned | Send briefs now; drop files into the narration manifest |

SECTION 23

# What's next

| # | Next step | Outcome it unlocks |
|---|---|---|
| 1 | Run the Week-1 device spikes on a Quest: passthrough capture, shadow look on a close and a far wall, recognition across 5+ testers, record silhouette templates | The go criteria in Section 12, and which shapes to keep, drop or loosen |
| 2 | Commission the voice actor and composer (with distribution rights) and run the script table read | Real narration and score in time for Week 5; the clip manifest is already waiting |
| 3 | Tune joint templates, silhouette thresholds and tolerances on the recorded data | At least 90% recognition with no false matches while hands move through the light |
| 4 | Set up hosting on HTTPS and keep the link live; later, PWA packaging for the Horizon Store | The submission link; installable build for testers |
| 5 | On-device performance pass toward 72 fps | Meets the 60 fps judging floor with margin |
| 6 | Playtests in 10+ real rooms, including desk-against-wall and no wall; first-five-minutes iteration | Every cold tester makes their first shape unprompted |
| 7 | Capture footage from Week 4 onward and cut the under-3-minute video | Submission video |
| 8 | Review and merge pull request #1 into main | A stable main branch to freeze for submission |

### Bottom line

The experience described in this brief exists and plays end to end in the emulator, with every must-ship item built except the commissioned narration and score. The remaining risk sits where the brief always said it would: recognition with real hands and how real the shadows look. Both need a headset, and they come first.

SECTION 24

# Bedtime Across the World

The original brief is a single-player shadow-puppet story. The build turns it into something families use every night: **a bedtime story that a parent tells with their hands, for a child who is far away**. A parent on deployment, working abroad or across an ocean tells the story in shadows on their wall in a Quest; the child watches it, plays along and answers on a phone or tablet at home. Everything in the original brief stays: it is now the instrument a parent plays, not only a game a child plays.

## What a family can do

| Mode | Parent (Quest) | Child (phone or tablet) |
|---|---|---|
| **Record and send** | Tells a story with hands and voice; Send to family gives a private three-word code. | Watches it in the parent's voice, with their own name in it; ends with a goodnight card. |
| **Join in** | Nothing extra. | At each shape the story waits; the child's own hands (camera, on device) wake the creature. |
| **Goodnight back** | Goodnights waiting in the headset play on the wall. | Twenty seconds of their shadow creature, and their voice if a grown-up allows. |
| **Live together** | Go live, read the code; Talk opens the microphone. | Joins with the code; sees the parent's shadows in real time; talks. |
| **Take turns** | Turns on Take turns. | Makes every other creature, on both screens. |

## A library of stories

| Story | About | Creatures |
|---|---|---|
| **The Moth Who Borrowed Light** | The original five-chapter adventure, with a branching wolf encounter and the lighthouse finale (about 20 min). | rabbit, fish, dog or bird, bird, cup |
| **Goodnight Across the Sea** | Mo carries a goodnight from someone far away to the child's window (5 min). | rabbit, fish, bird, cup |
| **The Letter That Took the Slow Way** | Waiting for someone: a letter that comes home slowly (5 min). | snail, swan, deer, cup |
| **The Brave Little Dog** | Being brave while someone is away (5 min). | dog (grows against the wolf), crocodile, bird |
| **Counting Stars** | A calm count to five for the littlest ones; the lamp dims as it goes (3 min). | bird, fish, cup |

- All five are narrated, and lines that greet the child show their name.
- Someone trying it alone gets a labelled **sample family**: a story from Dad on the viewer and a goodnight from Mia in the headset.

SECTION 25

# Build status (as of 10 Nov 2026)

Merged to **main** (pull requests #1 and #2) and live at **https://shadowplay-1031148889398.us-central1.run.app** on Cloud Run with a private Cloud Storage bucket. Typecheck, 13 server tests and the recognition sweep run in CI on every push.

| Area | Status | What exists |
|---|---|---|
| **Headset experience** | **Built** | Wall stage from planes with screen fallback, room dimming, draggable lamp, capsule hand shadows, dual recognition (joint angles and silhouettes), eleven shapes, assist ladder, Shadow Journal, Free Stage, accessibility settings. |
| **Story library** | **Built** | Five stories; a picker in the headset (browse, then Tell this story); stories are data (JSON) written by a script. |
| **Narration** | **Built** | 174 clips generated with Kokoro (open Apache-2.0 model) for every captioned line; silent while a parent narrates themselves. |
| **Record, share, watch** | **Built** | Performance stream (30 Hz hands, beats, events, voice), private family codes with 30-day expiry, family viewer, join-in with camera hands. |
| **Goodnights back** | **Built** | Child records up to 20 s; stored with the story's code; the headset finds and plays them. |
| **Live rooms** | **Built** | Real-time story and hands, voice both ways with an adaptive jitter buffer, duet turns, reconnects through dropouts (the room waits 45 s). |
| **Sample family** | **Built** | Story from Dad and goodnight from Mia, added once per device on first launch. |
| **Security** | **Built** | Protocol-only relay, rate limits on the real client address, strict story validation, room and message limits; covered by server tests. |
| **Hosting** | **Live** | Cloud Run (one instance, scales to zero), dedicated service account, deployed smoke test (21 checks). |
| **Submission video** | **Done** | 2:33, 1080p, from the running app and the emulator; original music; footage pipeline in the repo. |
| **Physical Quest** | **Not run** | No device available; see Section 27. |

SECTION 26

# Verified without a headset or a second person

Without a Quest or testers, every family-facing claim was checked by simulation: headless browsers, the IWSDK emulator, Chrome's fake camera and microphone (files, never real devices), a network impairment relay and bot playtesters. The scripts are in the repository and can be rerun.

## Two simulated families (live story)

| Check | Result |
|---|---|
| **Story steps reaching the child** | All, in order (63 of 63 in a full run); steps during a dropout arrive as a snapshot on reconnect |
| **Voice both ways** | About 99.6% of packets delivered under impairment |
| **Dropout** | Recovered in about 2.3 s; a parent who leaves for good ends the room after 45 s |
| **Take turns** | Turn passes to the child at a shape request and back after the creature bows (or if no hands appear) |
| **Ending** | The child's screen ends within 0.8 s when the parent ends the story |

## Voice delay (mouth to ear, voice-only lab)

| Network | Delay |
|---|---|
| **Same city** | 56 ms |
| **Abroad (120 ms one way, jitter)** | 336 ms |
| **Satellite or deployed (300 ms, heavy jitter)** | About 800 ms |
| **Back to clean after a bad patch** | 56 ms (the adaptive buffer recovers) |

## Bot playtesters (The Moth, 4 minutes each) and the new stories

| Player | Result |
|---|---|
| **Perfect, small hands (75%), shaky hands** | Finished the whole story; every shape recognised in about a second |
| **Wrong shape first** | Wrong shapes never woke the wrong creature; reached Chapter Five |
| **Slow (25 s per shape), wanderer (hands gone 30 s)** | Hints and the first assist step appeared; never stuck |
| **New stories (director)** | The Letter, The Brave Little Dog and Counting Stars all finished with no errors |
| **Performance (whole story)** | Peak 70 draw calls and 39k triangles; about 0.6 ms of app work per frame on desktop |

## Found and fixed by the simulations

- Voice delay that never recovered after a bad network patch (now adaptive).

- Instant story steps arriving together could overwrite each other (now one per frame).
- The recorder fell behind below 30 fps, so replayed hands drifted from the voice (now time-based).
- A guest could send a fake "story ended" through the relay (the server now relays protocol messages only).

SECTION 27

# Risks and gaps

| Gap | Why it is open | Mitigation |
|---|---|---|
| **Never run on a physical Quest** | No device available | Standard WebXR features with fallbacks (no planes: floating screen; no hands: pause and ask); performance far inside budgets in simulation |
| **Real two-person voice, real camera hands** | Tested only with fake devices | Echo cancellation and noise suppression from the browser; adaptive jitter buffer; the camera pipeline is the same MediaPipe model |
| **Commissioned narration** | Generated voice instead | Apache-2.0 voices; parents' own voices are the heart of the experience; clips can be replaced any time |
| **Scale beyond one server instance** | Live rooms are in memory | One instance is enough for judging; a shared room registry later |

## Mapping to the judging criteria

| Criterion | Evidence |
|---|---|
| **Innovation and Creativity** | An ancient art form, reinvented with hand tracking, becomes a way for families apart to share bedtime: recorded, joined in, answered and live. |
| **Experience Design** | Pre-learned gestures, no text tutorials, a story that waits and helps; one tap for a child to watch or join; safe by design. |
| **Technical Implementation** | Wall detection, real-time hand shadows, dual recognition, a replayable performance stream, live rooms with voice, all verified by simulation. |
| **Polish and Presentation** | Silhouette art, narrated library, a sample family for anyone alone, and a video that tells the family story. |

SECTION 28

# Submission

| Field | Entry |
|---|---|
| **Name** | Shadowplay: Bedtime Across the World |
| **Tagline (131 characters)** | Bedtime stories told with your hands, for a child far away: shadow puppets on your wall, watched, joined in and answered from home. |
| **Track / division** | Entertainment / New Experience |
| **Experience link** | https://shadowplay-1031148889398.us-central1.run.app |
| **Family viewer** | https://shadowplay-1031148889398.us-central1.run.app/?viewer |
| **Video** | shadowplay-video.mp4 (2:33), uploaded to YouTube or Vimeo before submitting |
| **Target launch** | Live now as a web beta; Horizon Store (PWA) in Q1 2027 with more stories |

The full description (about 500 words), the hand-interaction notes, how judges can try it and the rules checklist are in **docs/SUBMISSION.md**.

## Plan to the deadline (18 Nov, 12:00 PT)

- Final look at the video; upload it; fill in the Devpost form.
- Freeze on 17 Nov: tag the deployed commit and stop deploying; run the deployed smoke test once more.
- Keep the Google Cloud project and billing active until the winners are announced (about 11 Dec).




From the submission video: Dad's wall and Mia's tablet, the same story in sync.